Fraud Detection in Banking Transactions

Machine Learning Analysis and Modelling Report

Author	NDJOMNANG NJOMOU Daniel Alain Junior
Dataset	Credit Card Fraud Detection (Kaggle / ULB)
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Technology	Python · scikit-learn · XGBoost · imbalanced-learn

1. Introduction

Banking fraud poses a growing threat to financial institutions and their customers. With the rise of digital payments, fraudsters are developing increasingly sophisticated strategies to circumvent traditional security systems.

This project aims to build an automated fraud detection system using machine learning, capable of identifying suspicious transactions in near real time. To do this, we are using the public *'Credit Card Fraud Detection'* dataset made available by the Université Libre de Bruxelles (ULB).

2. Description of the Dataset

The dataset contains credit card transactions made in September 2013 by European cardholders. It covers two days' worth of transactions.

Characteristics	Value
Number of transactions	284,807
Fraudulent transactions	492 (0.17%)
Normal transactions	284,315 (99.83%)
Number of variables	31 (Time, V1–V28, Amount, Class)
Missing values	None
Variables V1–V28	PCA components (anonymised)

3. Data Exploration

The exploratory analysis highlighted two key characteristics of the dataset
: a significant imbalance between classes and a notable difference in the distribution of amounts.

3.1 Distribution of amounts

The graph below compares the distribution of amounts for normal transactions (in blue) and fraudulent transactions (in red). It can be seen that:

• Normal transactions reach up to €25,000, with a concentration between €0 and €1,000.

• Fraudulent transactions are predominantly below €500, indicating that fraudsters prefer small sums to avoid detection.

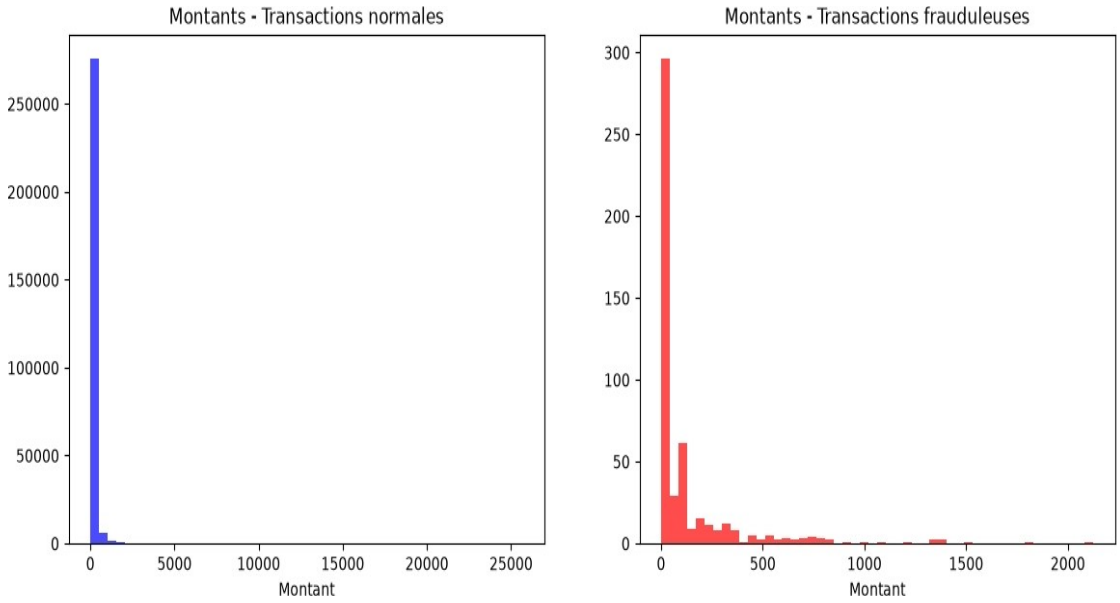

Figure 1 – Distribution of amounts by category

3.2 Correlation matrix

The correlation matrix reveals that variables V1 to V28 (derived from a PCA) are virtually independent of one another — which is ideal for ML algorithms. Only a few variables (V2, V4, V11) show a notable correlation with the target variable **Class**.

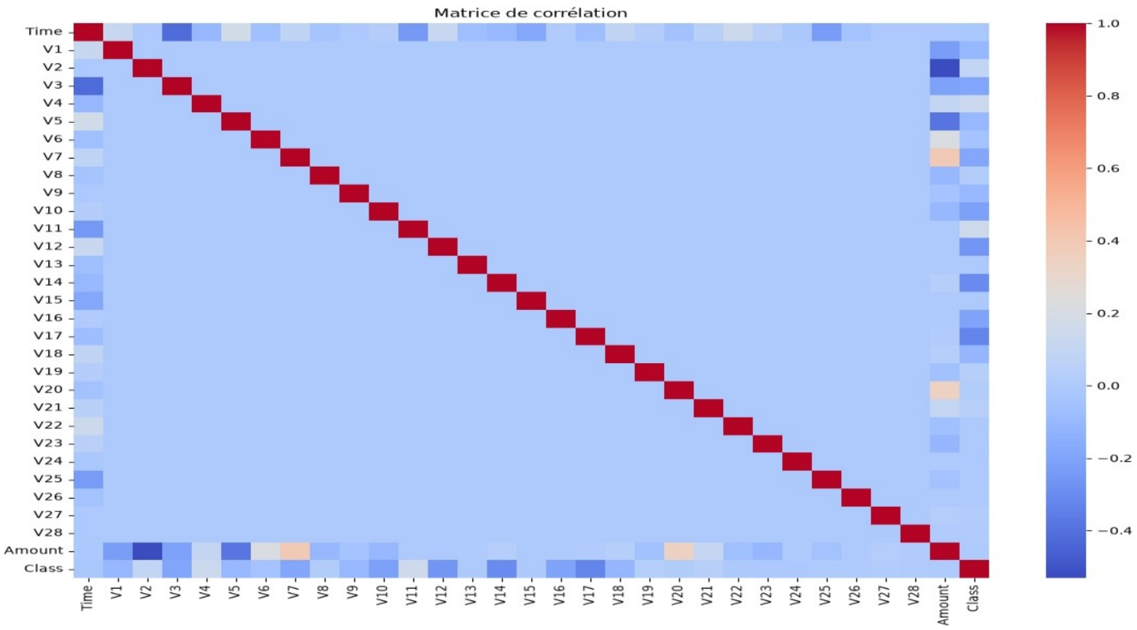

Figure 2 – Correlation matrix

4.Data Pre-processing

Before training the models, several pre-processing steps were carried out:

Step	Description
Normalisation	StandardScaler applied to the Time and Amount variables
Training/test split	70% training — 30% test (stratified)
Handling of imbalance	SMOTE: synthetic generation of fraudulent examples
Result after SMOTE	199,020 normal vs 199,020 fraudulent (balanced)

5. Machine Learning models

Five models were trained and compared. For each model, the key metrics are precision, recall and F1-score on the fraud class (Class=1).

5.1 Comparison table

Model	Precision	Recall	F1-score	False positives	AUC
Random Forest Base	0.87	0.79	0.83	18	0.954
XGBoost	0.75	0.80	0.78	39	0.974
Optimised Random Forest	0.78	0.80	0.79	33	0.964
Logistic Regression	0.06	0.88	0.12	1919	—
SVM (LinearSVC)	0.06	0.87	0.11	2008	—

5.2 Confusion matrices

Confusion matrices allow us to visualise true positives (correctly detected fraud), false positives (false alarms) and false negatives (missed fraud) for each model.

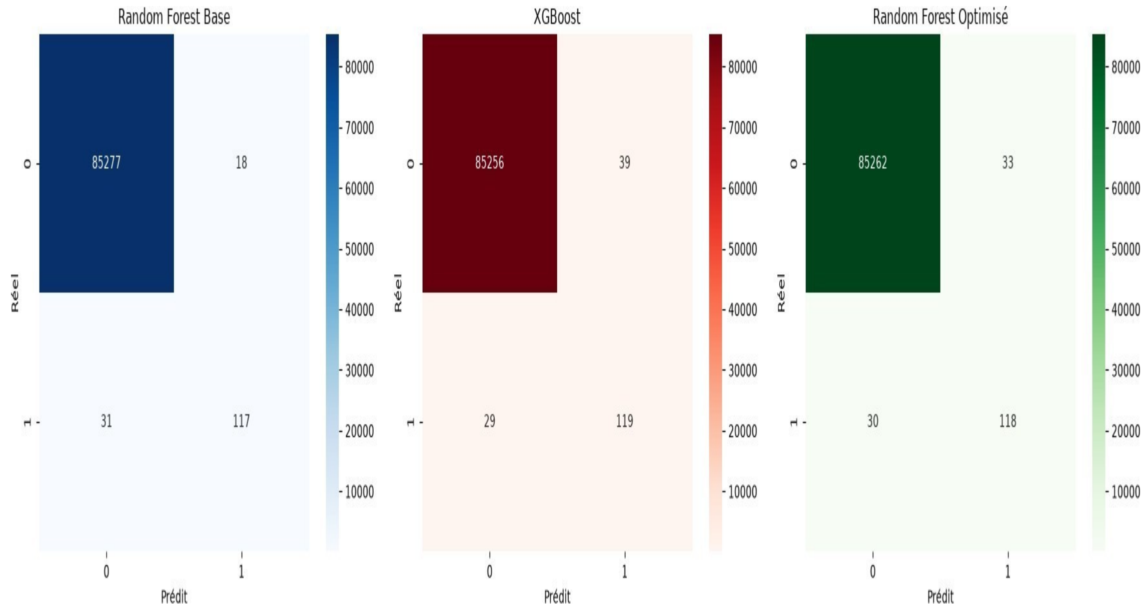

Figure 3 – Confusion matrices for the three best models

5.3 ROC curve

The ROC (Receiver Operating Characteristic) curve measures the model's ability to distinguish fraud from normal transactions. The AUC (Area Under the Curve) is better the closer it is to 1.

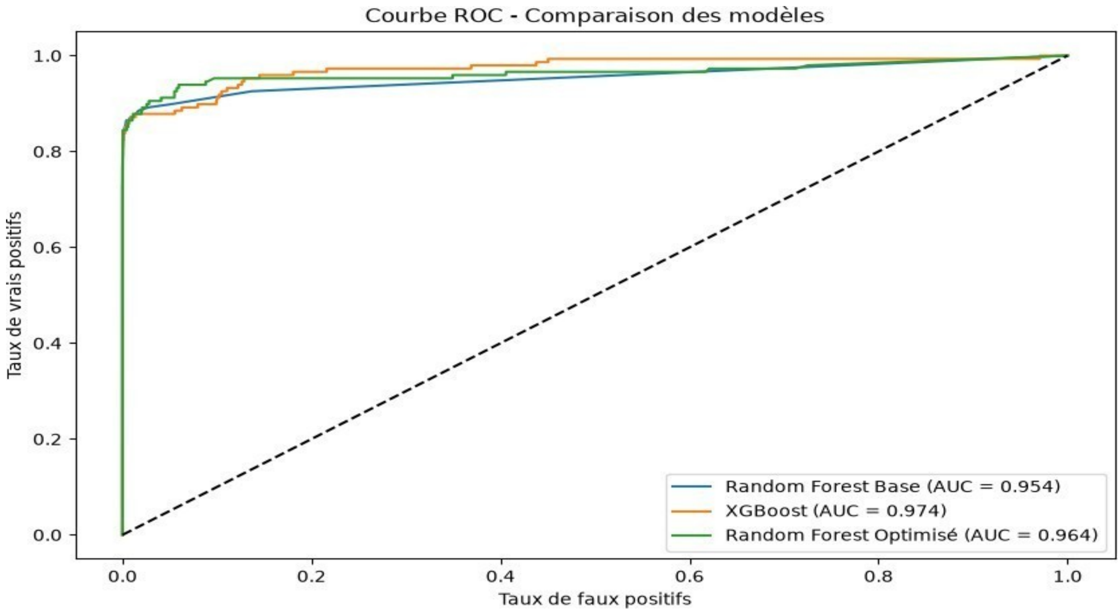

Figure 4 – ROC curve and AUC for the three best models

6. Interpretation and Conclusion

6.1 Analysis of the results

The results clearly show that ensemble models (Random Forest and XGBoost) significantly outperform linear models (Logistic Regression and SVM) for this fraud detection problem.

Whilst Logistic Regression and SVM achieve high recall rates (88% and 87% respectively), they generate a staggering number of false positives (1,919 and 2,008 respectively). In practice, this would mean blocking thousands of legitimate transactions every day, which is unacceptable in a production environment.

6.2 Best model

XGBoost is the recommended model with an AUC of 0.974 — the highest of all the models tested. It offers the best balance between fraud detection (recall = 0.80) and minimising false alarms (only 39).

The basic Random Forest remains an excellent alternative, with the best F1-score (0.83) and the fewest false alarms (18), albeit at the cost of a slightly lower AUC.

6.3 Areas for improvement

Several avenues can be explored to further improve performance:

• Optimisation of XGBoost hyperparameters using Pipeline + RFECV

• Adding new variables (feature engineering)

• Use of neural networks (LSTM, Autoencoder) to detect anomalies

• Deploying the model via a REST API (Flask/FastAPI) or a Streamlit interface

• Implementing an automatic retraining system (drift detection)

7. Conclusion

This project demonstrates the effectiveness of machine learning algorithms for detecting banking fraud. Thanks to a rigorous methodology — data pre-processing, handling class imbalance using SMOTE, comparing five models and optimising hyperparameters — we have developed a system capable of detecting 80% of fraud cases with a very low false alarm rate.

XGBoost, with an AUC of 0.974, is the model selected for potential deployment in production. This work provides a solid foundation for a real-time banking transaction monitoring system.

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